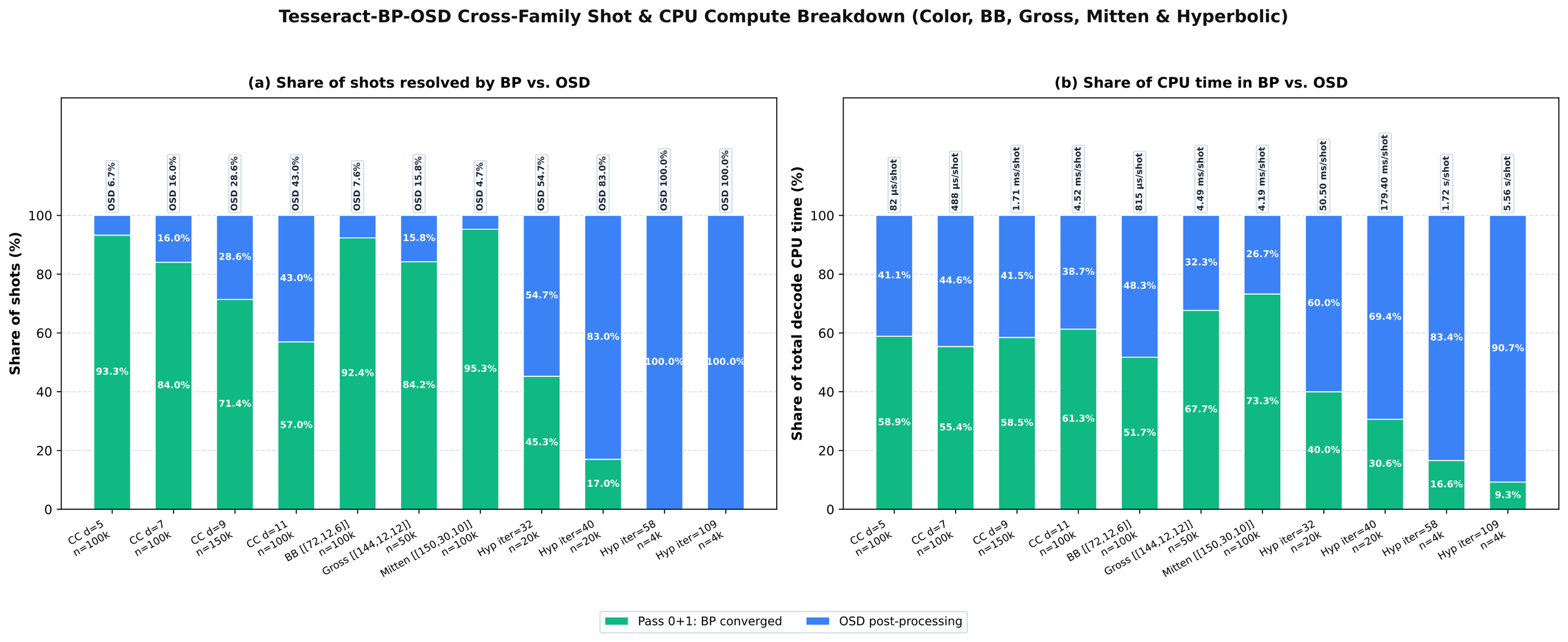
Tesseract-BP-OSD Cross-Family Shot & CPU Compute Breakdown (Color, BB, Gross, Mitten & Hyperbolic)
(a) Share of shots resolved by BP vs. OSD
Share of shots (%)
0
20
40
60
80
100
OSD 6.7%
OSD 16.0%
OSD 28.6%
OSD 43.0%
OSD 7.6%
OSD 15.8%
OSD 4.7%
OSD 54.7%
OSD 83.0%
OSD 100.0%
OSD 100.0%
93.3%
84.0%
71.4%
57.0%
92.4%
84.2%
95.3%
45.3%
17.0%
100.0%
100.0%
CC d=5 n=100k
CC d=7 n=100k
CC d=9 n=150k
CC d=11 n=100k
BB [[72,12,6]] n=100k
Gross [[144,12,12]] n=50k
Mitten [[150,30,10]] n=100k
Hyp iter=32 n=20k
Hyp iter=40 n=20k
Hyp iter=58 n=4k
Hyp iter=109 n=4k
(b) Share of CPU time in BP vs. OSD
Share of total decode CPU time (%)
0
20
40
60
80
100
82 µs/shot
488 µs/shot
1.71 ms/shot
4.52 ms/shot
815 µs/shot
4.49 ms/shot
4.19 ms/shot
50.50 ms/shot
179.40 ms/shot
1.72 s/shot
5.56 s/shot
41.1%
44.6%
41.5%
38.7%
48.3%
32.3%
26.7%
60.0%
69.4%
83.4%
90.7%
58.9%
55.4%
58.5%
61.3%
51.7%
67.7%
73.3%
40.0%
30.6%
16.6%
9.3%
CC d=5 n=100k
CC d=7 n=100k
CC d=9 n=150k
CC d=11 n=100k
BB [[72,12,6]] n=100k
Gross [[144,12,12]] n=50k
Mitten [[150,30,10]] n=100k
Hyp iter=32 n=20k
Hyp iter=40 n=20k
Hyp iter=58 n=4k
Hyp iter=109 n=4k
Pass 0+1: BP converged
OSD post-processing